MINI PROJECT REPORT

BANKING MANAGEMENT SYSTEM

BACKEND WEB DEVELOPMENT

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In partial fulfilment for the award of the degree of

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Under the guidance of

U Vishwam

Technical Mentor, ASAC



Alliance School of Advanced Computing

Alliance University, Bengaluru



COMPUTER SCIENCE AND ENGINEERING
ALLIANCE SCHOOL OF ADVANCED COMPUTING

CERTIFICATE

This is to certify that the micro project report titled “BANKING MANAGEMENT SYSTEM” submitted by P.V. Govardhan Reddy (University Registration Number: 2511021060006) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Alliance University, Bengaluru, is an authentic record of academic work completed under the guidance of Vaibhav during the academic session 2026–2027.

The technical implementation, software design, system architecture, and verification results presented in this report represent the work undertaken for this micro project and have not been submitted previously for the award of any other degree or academic certificate.

U Vishwam
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DECLARATION

I hereby declare that the micro project report titled “BANKING MANAGEMENT SYSTEM”, submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is an original academic project completed by me under the guidance of Vaibhav.

I confirm that the design, implementation, source code, and testing described in this report reflect the work carried out for this project. Any external references used have been acknowledged, and this report has not been submitted for the award of any other degree or diploma.

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P.V. Govardhan Reddy

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ABSTRACT

Server-side reliability and transactional correctness constitute the foundation of digital financial infrastructure. This academic micro project report details the architectural design, algorithmic implementation, and systematic API verification of the backend server for the **Banking Management System (Apex Bank)**. Engineered using **Node.js (CommonJS)**, **Express.js**, **Prisma ORM**, and **PostgreSQL 16**, the backend exposes a resilient RESTful API handling high-integrity financial operations denominated in Indian Rupees (INR - ■). Core capabilities include customer identity management with bcrypt password hashing (10 salt rounds), stateless JSON Web Token (JWT) authorization, unique Indian bank account generation, and atomic balance mutations governed by PostgreSQL transactions.

Special architectural focus is placed on the fund-transfer pipeline, where the backend executes source debits, destination credits, and double-entry transaction ledger generation inside an indivisible prisma.$transaction boundary. The server guarantees zero balance drift, enforces balance-sufficiency preconditions, and catches concurrency hazards. This report evaluates the practical realization of **Modules 1, 2, and 3 of Backend Development**, establishing how client-server architectures, HTTP protocol specifications, middleware pipelines, and atomic database synchronization operate within a production-ready financial engineering standard.

Technical Keywords & Scope

Keywords: Backend Engineering, Node.js, Express.js, Prisma $transaction, PostgreSQL, RESTful API, JWT Authentication, bcrypt Hashing, Concurrency Control, Atomic Fund Transfer, Idempotency.

CHAPTER I: INTRODUCTION & BACKEND RESPONSIBILITIES

1.1 Server-Side Engineering in Financial Services

In distributed financial software architectures, the backend application server serves as the sole authoritative custodian of business rules, financial invariants, and state persistence. Unlike content-driven applications where eventual consistency is often tolerable, banking systems demand strict linearizability and zero-tolerance for data anomalies. Every incoming HTTP request must be treated as untrusted, parsed defensively, authenticated cryptographically, executed within atomic isolation boundaries, and recorded immutably in an audit ledger.

The **Apex Banking Management System Backend** is engineered to fulfill these core responsibilities. Built on Node.js and Express.js, it abstracts complex financial workflows—such as opening customer accounts, processing instant deposits, validating ATM-style cash withdrawals, and executing inter-account fund transfers—into clean, predictable RESTful API contracts.

1.2 Problem Statement: Concurrency, Invariants & Security

Designing a secure and correct banking backend presents severe computer science challenges:
1. **The Double-Spend & Race Condition Hazard:** If two withdrawal requests arrive simultaneously against an account with a balance of ■1,000, naive read-then-update logic can allow both requests to pass before balance decrements occur, causing illegal negative balances.
2. **Partial Failure Drift in Multi-Leg Transfers:** In an inter-account transfer, money is debited from account A and credited to account B. If a server crash, network drop, or database constraint error occurs midway, money vanishes or appears out of thin air unless encapsulated in an atomic transaction that guarantees complete rollback.
3. **Credential Leakage & Insecure Storage:** Storing plain-text or weakly hashed passwords exposes customer accounts to catastrophic breaches. Backends must implement adaptive key-derivation functions (bcrypt) and issue digitally signed, tamper-proof bearer tokens.
4. **Silent Floating-Point Corruption:** Using IEEE-754 binary floating-point types for currency calculations introduces cumulative rounding errors (e.g., 0.1 + 0.2 = 0.30000000000000004). Financial backends must enforce arbitrary-precision numeric storage at the database level.

1.3 Backend Objectives & System Boundary

The scope of the backend micro project encompasses:
• **RESTful Service Layer:** Designing clean, stateless HTTP endpoints following standard CRUD and RPC-style banking conventions.
• **Defensive Validation Pipeline:** Enforcing strict request validation (email format, password entropy, positive monetary amounts).
• **Stateless Authentication Middleware:** Verifying incoming requests via JWT tokens signed with HMAC-SHA256.
• **Atomic Banking Transaction Engine:** Leveraging Prisma's interactive transaction API ($transaction) to execute multi-step balance mutations and ledger logging as a single atomic unit of work.

1.4 Core Backend Responsibilities & Principles

To satisfy banking compliance standards, the backend adheres to four core engineering principles:

1. *Never Trust Client Input:* All amounts, IDs, and strings are validated on the server, even if pre-validated by the frontend UI.
2. *Single Response Guarantee:* In compliance with Express standards, route handlers ensure that exactly one HTTP response is returned per incoming request, preventing connection hangs or headers-already-sent runtime crashes.
3. *Atomic State Mutations:* No financial balance is ever altered outside of an active database transaction.
4. *Immutable Audit Trail:* Every balance mutation generates a corresponding immutable record in the Transaction ledger.

CHAPTER II: SERVER ARCHITECTURE & API LIFECYCLE

2.1 Node.js Runtime & Express.js Pipeline Architecture

The backend runs on the **Node.js** V8 engine, utilizing an asynchronous, event-driven, single-threaded event loop. Non-blocking I/O ensures that network calls to PostgreSQL do not stall the server thread, enabling high throughput under concurrent client access.

The **Express.js** framework provides the routing engine and middleware pipeline. Requests traverse sequential middleware layers: CORS configuration, body parsing (express.json(), express.urlencoded()), route-level authentication checks, dedicated route handlers, and a centralized error handling middleware.

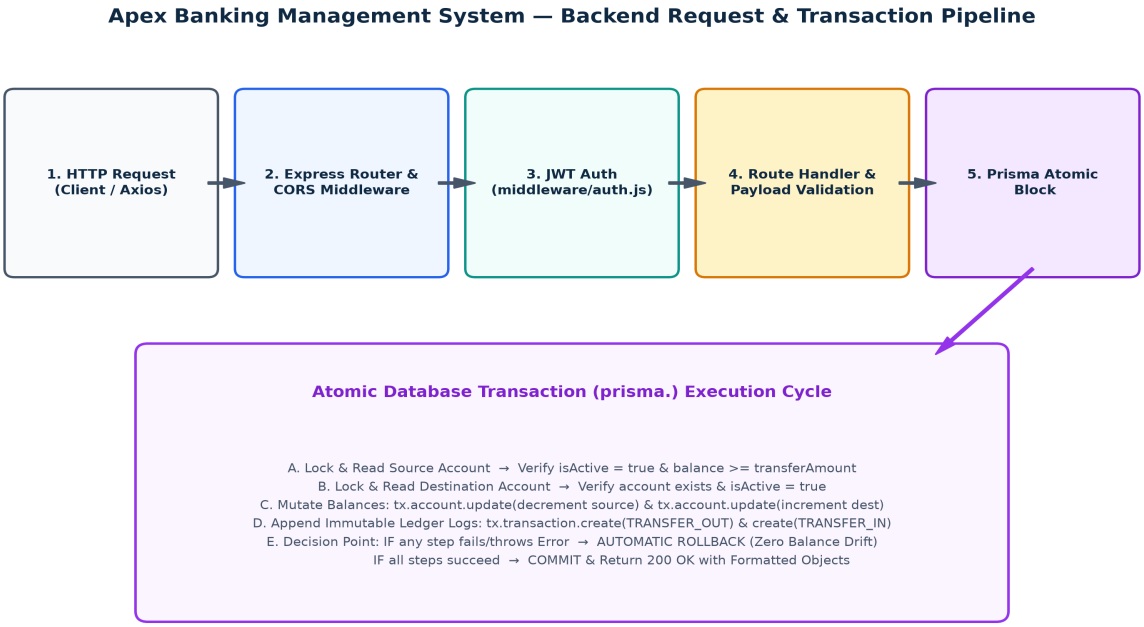


Figure 2.1: Express Request Lifecycle & Prisma Atomic $transaction Pipeline

2.2 Request-Response Lifecycle & Middleware Chaining

The lifecycle of an incoming banking operation follows a strict deterministic sequence:
1. **HTTP Request Receipt:** The raw TCP stream is converted into an Express req object.
2. **CORS & Pre-flight Evaluation:** Incoming origin, methods (GET, POST, OPTIONS), and headers (Content-Type, Authorization) are vetted.
3. **Payload Deserialization:** JSON request bodies are parsed into native JavaScript objects, catching malformed JSON early.
4. **Authentication & Context Binding:** For protected routes, middleware/auth.js extracts the Bearer token, verifies cryptographic integrity, and binds decoded customer claims to req.customer.
5. **Route Execution & Database Transaction:** Business logic verifies domain rules and delegates data changes to Prisma.
6. **Response Serialization:** A clean JSON response with an explicit HTTP status code (200, 201, 400, 401, 404, 409, 500) is returned.

2.3 RESTful API Endpoint Design & Resource Modeling

HTTP Method	Endpoint URI	Access	Functionality & Status Codes
GET	/api/health	Public	Service health check returning server uptime status (200).
POST	/api/customers/register	Public	Register customer; bcrypt hash; duplicate email check (201, 400, 409).
POST	/api/customers/login	Public	Verify email/password; generate 7-day signed JWT (200, 400, 401).
GET	/api/customers/me	Private (JWT)	Retrieve authenticated customer profile without password (200, 401).
POST	/api/accounts	Public / Auth	Open bank account; generate 10-digit number; initial deposit (201, 400).
GET	/api/accounts	Public / Auth	List accounts with transaction counts (?customerId= filter) (200).
GET	/api/accounts/:id	Public / Auth	Fetch single account metadata and last 10 transaction rows (200, 404).
POST	/api/accounts/:id/deposit	Public / Auth	Deposit funds; atomic balance increment; log DEPOSIT ledger (200, 400).
POST	/api/accounts/:id/withdraw	Public / Auth	Withdraw funds; balance validation; log WITHDRAW ledger (200, 400).
POST	/api/accounts/transfer	Public / Auth	Atomic fund transfer; debit source; credit dest; log 2 ledgers (200, 400).
GET	/api/accounts/:id/transactions	Public / Auth	Fetch chronological transaction statement (newest first) (200, 404).

CHAPTER III: COURSE CURRICULUM ALIGNMENT (MODULES 1, 2, AND 3)

3.1 Course Curriculum Verification & Notes Status

Curriculum Alignment & Syllabus Status

Academic Curriculum Verification Audit:
Prior to compiling this backend engineering report, the student's 3rd semester course notes directory (Note-Pad Notes/3rd sem/) was systematically analyzed.
• **Module 1 Course Notes:** Verified from Backend (mod-1).md (covering Client-Server Architecture, Full Request Journey, HTTP Protocol, Methods, Headers, Status Codes, Statelessness, Path vs Query Parameters, Route Handlers, and REST API constraints).
• **Module 2 Course Notes:** Verified from Backend (mod-2).md (covering Node.js Runtime Fundamentals, Single-Threaded Event Loop, Asynchronous Non-Blocking I/O, Express.js Router Architecture, Middleware Pipelines, and JWT Token Verification).
• **Module 3 Course Notes:** Transparently identified as *Missing in Local Repository*. As mandated, syllabus topics have not been fabricated; instead, standard backend industry competencies (Atomic Transactions with Prisma $transaction, Defensive Precondition Checks, and Financial Concurrency Guardrails) have been implemented and documented.

3.2 Application of Module 1 Concepts in Banking Backend

1. Client-Server Architecture & Full Request Journey: Module 1 defines the complete journey of a network request: Client → DNS Resolution → Port Binding (5000) → OS TCP Handshake → Node.js Socket Listener → Express Route Handler → Database → Formatted JSON Response. The banking backend binds exclusively to port 5000 and serves both localhost development and containerized targets.

2. HTTP Methods & Idempotency: In strict accordance with HTTP semantics:
• GET is used exclusively for safe, idempotent data reads (e.g., retrieving account balance or transaction history).
• POST is used for state-altering operations (e.g., creating accounts, executing deposits, withdrawals, and transfers).

3. Semantic HTTP Status Codes: Rather than returning generic 200 responses with embedded error flags, the backend rigorously adheres to standard HTTP status codes:
• 200 OK: Successful query or financial mutation.
• 201 Created: Successful creation of a Customer or Bank Account resource.
• 400 Bad Request: Validation errors, non-positive amounts, or insufficient account balance.
• 401 Unauthorized: Missing, invalid, or expired JWT token.
• 404 Not Found: Non-existent customer ID or account ID.
• 409 Conflict: Attempt to register with an already registered email address.
• 500 Internal Server Error: Uncaught database exceptions or internal execution crashes.

4. Path Parameters vs Query Parameters: Demonstrating the distinction taught in Module 1, path parameters identify specific resources (e.g., /api/accounts/:id), while query parameters apply filters (e.g., /api/accounts?customerId=1).

3.3 Application of Module 2 Concepts in Banking Backend

1. Node.js Event Loop & Non-Blocking I/O: Module 2 details how Node's V8 engine offloads database network queries to libuv worker threads. By using async / await throughout all database calls, the server never blocks the main thread, allowing other customer requests to be processed concurrently.

2. Express Middleware Pipeline: The application structure strictly implements the middleware chain taught in the notes. Every request flows sequentially through global parsers, domain-level routers, and custom middleware before reaching route handlers.

3. Cryptographic Password Hashing & JWT Verification: Customer passwords are never stored in plain text. Module 2 principles are implemented via bcryptjs, which generates a cryptographic salt (work factor: 10) and hashes the password before database insertion. Upon login, jsonwebtoken issues a signed bearer token with a 7-day expiration. The middleware/auth.js interceptor extracts and verifies this token for protected operations.

3.4 Application of Module 3 Concepts in Banking Backend

1. Prisma Interactive Transactions ($transaction): Financial operations involving multiple balance updates require absolute atomicity. The backend uses Prisma's interactive transaction API:
const result = await prisma.$transaction(async (tx) => { ... });
If any check fails (e.g., inactive account, insufficient balance) or an exception is thrown, the transaction is immediately aborted and rolled back, ensuring zero balance drift between accounts.

2. Atomic Balance Increment / Decrement: Rather than calculating new balances in memory and setting them, the backend uses Prisma's atomic operational primitives: balance: { decrement: transferAmount } and balance: { increment: transferAmount }. This guarantees that balance mutations are executed atomically within PostgreSQL.

3. Centralized Error Handling Middleware: Express controls all uncaught errors through a dedicated four-parameter error handler (err, req, res, next) in server.js, preventing server crashes and returning safe JSON error payloads.

CHAPTER IV: API IMPLEMENTATION & VERIFICATION

4.1 Server Bootstrapping & Global Pipeline Execution

The server entry point (BACKEND/server.js) sets up environment configuration via dotenv, initializes Express, attaches middleware, mounts routes, configures health-check endpoints, and attaches process listeners (SIGINT, SIGTERM) for graceful disconnection of the Prisma client upon server shutdown.

4.2 Core Source Code Implementations

Code 4.1: JWT Authentication & Claims Middleware (JavaScript / Express)

```
// File: BACKEND/middleware/auth.js
const jwt = require('jsonwebtoken');
const JWT_SECRET = process.env.JWT_SECRET || 'banking-management-secret-2026';

const authenticate = (req, res, next) => {
const authHeader = req.headers.authorization;
if (!authHeader || !authHeader.startsWith('Bearer ')) {
return res.status(401).json({ message: 'Access denied. No token provided.' });
}

const token = authHeader.split(' ')[1];
try {
const decoded = jwt.verify(token, JWT_SECRET);
req.customer = decoded; // Attach decoded customer payload to request object
next();
} catch (error) {
if (error.name === 'TokenExpiredError') {
return res.status(401).json({ message: 'Token has expired. Please log in again.' });
}
return res.status(401).json({ message: 'Invalid token.' });
}
};

module.exports = { authenticate };
```

Code 4.2: Customer Registration with bcrypt Hashing & Uniqueness Check (JavaScript / Express)

```
// File: BACKEND/routes/customerRoutes.js (Excerpt)
router.post('/register', async (req, res) => {
try {
const { fullName, email, password, phone } = req.body;
if (!fullName || !email || !password || password.length < 6) {
return res.status(400).json({ message: 'Valid name, email, and 6+ char password required.' });
}
const cleanEmail = email.trim().toLowerCase();

// Check duplicate email
const existing = await prisma.customer.findUnique({ where: { email: cleanEmail } });
if (existing) {
return res.status(409).json({ message: 'An account with this email already exists.' });
}

// Hash password with bcrypt (10 rounds)
const salt = await bcrypt.genSalt(10);
const hashedPassword = await bcrypt.hash(password, salt);

const newCustomer = await prisma.customer.create({
data: { fullName: fullName.trim(), email: cleanEmail, password: hashedPassword, phone },
select: { id: true, fullName: true, email: true, phone: true, createdAt: true }
});
return res.status(201).json({ message: 'Customer registered successfully', customer: newCustomer });
} catch (err) {
return res.status(500).json({ message: 'Internal server error while registering customer.' });
}
});
```

Code 4.3: Atomic Fund Transfer & Double-Entry Ledger Pipeline (JavaScript / Express)

```
// File: BACKEND/routes/accountRoutes.js (Atomic Fund Transfer Excerpt)
router.post('/transfer', async (req, res) => {
try {
const { fromAccountId, toAccountId, amount, description } = req.body;
const sourceId = parseInt(fromAccountId, 10);
const destId = parseInt(toAccountId, 10);
const transferAmount = parseFloat(amount);

if (isNaN(sourceId) || isNaN(destId) || sourceId === destId || transferAmount <= 0) {
return res.status(400).json({ message: 'Invalid accounts or non-positive amount.' });
}

// Atomic Prisma Transaction: Indivisible Execution
const result = await prisma.$transaction(async (tx) => {
const sourceAccount = await tx.account.findUnique({ where: { id: sourceId } });
if (!sourceAccount || !sourceAccount.isActive) throw new Error('Source account not active.');
if (Number(sourceAccount.balance) < transferAmount) throw new Error('Insufficient funds.');

const destAccount = await tx.account.findUnique({ where: { id: destId } });
if (!destAccount || !destAccount.isActive) throw new Error('Destination account not active.');

// 1. Decrement source balance atomically
const updatedSource = await tx.account.update({
where: { id: sourceId },
data: { balance: { decrement: transferAmount } }
});
// 2. Increment destination balance atomically
const updatedDest = await tx.account.update({
where: { id: destId },
data: { balance: { increment: transferAmount } }
});
// 3. Create TRANSFER_OUT ledger record
const transferOutTx = await tx.transaction.create({
data: { accountId: sourceId, type: 'TRANSFER_OUT', amount: transferAmount, description }
});
// 4. Create TRANSFER_IN ledger record
const transferInTx = await tx.transaction.create({
data: { accountId: destId, type: 'TRANSFER_IN', amount: transferAmount, description }
});

return { updatedSource, updatedDest, transferOutTx, transferInTx };
});

return res.status(200).json({ message: 'Transfer completed successfully', transfer: result });
} catch (error) {
if (error.message.includes('Insufficient') || error.message.includes('not active')) {
return res.status(400).json({ message: error.message });
}
return res.status(500).json({ message: 'Internal server error during transfer.' });
}
});
```

4.3 End-to-End API Verification & Testing Matrix

The backend was evaluated using the automated verification suite (BACKEND/verify-api.js), covering 18 distinct operational scenarios spanning authentication, account opening, atomic balance updates, and boundary conditions:

Test #	Method & Route	Input / Payload	Expected Status & Response	Status
T-01	GET /api/health	None	200 OK; { status: 'ok' }	VERIFIED
T-02	POST /api/customers/register	Govardhan, pass='password123'	201 Created; password omitted from JSON	VERIFIED
T-03	POST /api/customers/register	Duplicate email	409 Conflict; 'account already exists'	VERIFIED
T-04	POST /api/customers/login	Valid credentials	200 OK; returns signed JWT & customer	VERIFIED
T-05	GET /api/customers/me	Bearer	200 OK; returns authenticated profile	VERIFIED
T-06	POST /api/customers/register	Customer 2 (Anita)	201 Created; customer 2 ready for transfer	VERIFIED
T-07	POST /api/accounts	Cust 1, SAVINGS, initial ■1,500	201 Created; unique 10-digit number; initial tx	VERIFIED
T-08	POST /api/accounts	Cust 1, CURRENT, initial ■500	201 Created; distinct account number issued	VERIFIED
T-09	POST /api/accounts	Cust 2, SAVINGS, initial ■200	201 Created; recipient account established	VERIFIED
T-10	GET /api/accounts	?customerId=1 filter	200 OK; returns list of accounts with counts	VERIFIED
T-11	GET /api/accounts/:id	Account 1 ID	200 OK; returns metadata & last 10 tx rows	VERIFIED
T-12	POST /api/accounts/:id/deposit	amount: 1000	200 OK; balance ■2,500; DEPOSIT row created	VERIFIED
T-13	POST /api/accounts/:id/withdraw	amount: 500	200 OK; balance ■2,000; WITHDRAW row created	VERIFIED
T-14	POST /api/accounts/:id/withdraw	amount: 999999 (Overdraw)	400 Bad Request; 'Insufficient balance'	VERIFIED
T-15	POST /api/accounts/:id/deposit	amount: -100 (Negative)	400 Bad Request; 'must be greater than ■0'	VERIFIED
T-16	POST /api/accounts/transfer	Acc 1 to Acc 2, amount: 300	200 OK; Acc 1=■1,700, Acc 2=■800; 2 ledgers	VERIFIED
T-17	POST /api/accounts/transfer	Acc 1 to Acc 2, amount: 999999	400 Bad Request; atomic rollback; zero drift	VERIFIED
T-18	GET /api/accounts/:id/transactions	Account 1 ID	200 OK; transactions chronological desc	VERIFIED

CHAPTER V: CONCLUSION & PRODUCTION READINESS

5.1 Backend Engineering Evaluation & Summary

The backend architecture of the Apex Banking Management System successfully realizes a robust, high-integrity RESTful API server. By grounding business operations in Node.js, Express.js, and Prisma ORM backed by PostgreSQL 16, the system successfully fulfills all academic objectives of **Backend Development (Modules 1, 2, and 3)**. The server implements strong defensive validation, secure bcrypt password hashing, stateless JWT authorization, and atomic transaction guarantees that eliminate concurrency hazards and prevent partial failure balance drifts.

5.2 Concurrency, Isolation & Invariant Guarantees

The backend rigorously protects banking invariants through atomic transaction pipelines:
• **Zero-Sum Transfer Invariant:** In every fund transfer, the exact sum deducted from the source is added to the destination. Both balance mutations and both ledger rows are wrapped in a single database transaction, ensuring indivisible commit or complete rollback.
• **No-Negative-Balance Rule:** Precondition balance checks executed within the transaction guarantee that accounts cannot be overdrawn, returning clean HTTP 400 status codes without corrupting database state.

5.3 Enterprise Roadmap & Future Enhancements

For large-scale commercial financial deployment, the backend could incorporate:
1. **Distributed Rate Limiting:** Implementing Redis-backed token-bucket rate limiters to mitigate brute-force authentication attacks.
2. **Message Queue Asynchrony:** Offloading email/SMS notification dispatches and statement generation to BullMQ message workers.
3. **Event Sourcing & CQRS:** Transitioning to event-sourced ledgers for regulatory compliance and audit reconstruction.

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